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REVIEW

Experiencing financial toxicity associated with cancer in publicly funded healthcare systems: a systematic review of qualitative studies

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Abstract

Purpose Understanding how patients and families experience, respond to, and cope with the financial burden associated with cancer could assist in identifying future research priorities and developing relevant interventions to assist patients and families facing financial hardship. This systematic review offers a synthesis of the qualitative evidence on cancer-related financial toxicity from the perspective of patients and/or informal caregivers in publicly funded healthcare systems where it might be expected that financial protection would be strongest.

Methods Articles published between January 1, 2005, and March 7, 2019, describing financial burden experienced by cancer patients and/or informal caregivers were identified using OVID MEDLINE Embase and PsychInfo, CINAHL, Business Source Complete, and EconLit databases. English language, peer-reviewed qualitative papers describing studies conducted in countries with predominantly publicly funded healthcare systems were eligible. Quality appraisal was conducted using CASP Quality Appraisal Checklist. Narrative synthesis was completed with extracted data and themes identified inductively by all team members.

Results Twelve articles were identified as eligible. Articles reported on 10 studies conducted in Australia ($n = 2$), Canada ($n = 2$), England ($n = 3$), and Ireland ($n = 3$). The papers illustrate the complexity and multifaceted nature of experiencing financial hardship following a cancer diagnosis. Each contributes to the whole picture, providing different viewpoints regarding various and diverse forms of financial hardship, the process of confronting financial challenges, working to overcome difficulties, and coping with the resulting impacts. Synthesis of the studies suggested five themes: household and medical costs are increased, financial resources are reduced, financial change and financial hardship vary, financial hardship has many consequences, various mitigation strategies are used.

Conclusion Cancer patients and their families can experience a broad range of impacts when they are facing and coping with financial toxicity. Consistent gaps in support highlight that strategies to mitigate financial effects related to travel, accommodation, medications, family support, and income replacement are needed for many patients and families even in the context of publicly funded healthcare systems.

Implications for Cancer Survivors Survivors need to be informed early in their cancer experience about the potential financial burden associated with cancer treatment and its impact on survivors and their family members.

Keywords Financial toxicity · Cancer · Patient perspectives · Survivorship · Economic impact · Publicly funded Healthcare systems

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Introduction

Cancer and its treatment have physical, psychosocial, and practical consequences for individuals and their families [1]. The consequences may be felt before the actual diagnosis and continue to unfold throughout the treatment and follow-up periods. Conceptualization of cancer as an acute episode is shifting to embrace notions about its chronic nature [2] and the potential for a variety of impacts to be felt after the completion of primary treatment [3].

One practical consequence that has gained attention over the past decade is the financial burden on patients and their families that can be associated with a diagnosis of cancer [4, 5]. The term financial toxicity has been used to describe the distress and hardship from the financial burden of cancer [6]. This has been reported with cancer to greater or lesser degrees across all types of healthcare systems including the so-called publicly funded systems [7, 8]. It can manifest in a number of ways. Individuals can incur out-of-pocket costs, defined as direct expenditures related to cancer and its treatment that are not covered/reimbursed by government or insurers. Direct costs also include those for travel/parking and accommodation related to accessing treatment, medications, medical devices/supplies, and health professional services. Indirect costs to patients and their families, including income losses, can also be substantial [4, 5, 9, 10].

Changes in cancer care have heightened the relevance of better understanding the financial impact. In addition to improved survival rates and increased numbers of survivors, these include the high costs of newer cancer therapies, such as biological therapies [11], higher rates of healthcare debt for patients and families in some countries [12], and patients' decisions to forego treatment due to financial barriers [13]. These can all be exacerbated by the inability to work for significant periods of time during and following cancer treatment for both patients [14, 15] and their family members [16]. Survey data suggest associations between financial hardship and emotional distress and quality of life [17, 18].

Several reviews of financial toxicity have been completely focused largely on quantitative studies [4, 17–19]. Systematically identifying what has been uncovered through qualitative studies would be beneficial to deepen our understanding of the experience of financial toxicity for patients and their families. Understanding more about how patients and families experience, respond to, and cope with this consequence of cancer could assist in identifying future research priorities and developing relevant interventions to assist patients and families facing financial hardship in the future. We aimed, therefore, to conduct a systematic review and synthesis of the qualitative evidence describing cancer-related financial toxicity or burden experienced by patients and/or informal caregivers from their perspectives.

Methods

This review was completed as part of a larger systematic review designed to identify studies which reported on four categories of financial burden (i.e. out-of-pocket costs, income loss for patients and family caregivers, travel and accommodation costs, and patient perceived stress/strain) in publicly funded healthcare systems. Detailed rationale and description of the larger project together with findings from the systematic identification of quantitative literature is reported elsewhere [20].

In brief, articles published between January 1, 2005, and March 7, 2019, describing financial toxicity/burden experienced by cancer patients and/or informal caregivers, were identified using Ovid MEDLINE Embase, PsychInfo, CINAHL, Business Source Complete, and EconLit databases. Criteria for inclusion included English language peer-reviewed full papers describing studies conducted in countries with predominantly publicly funded healthcare (very high development index countries as per the United Nations Human Development Program 2018 listing, excluding the USA) for patients with one of the ten most common adult tumour sites (i.e. lung, breast, prostate, colorectal, bladder, cervix uteri, stomach, oesophagus, thyroid, liver) [21]. The term publicly funded healthcare can take a variety of forms including a National Health Service model, part of a social security system, or a mix of public and private systems. We included countries where the dominant system is a public one regardless of model type.

The original search produced 7117 unique citations. Screening for eligibility was first completed for titles (split across team members), followed by review of abstracts and review of full manuscripts; both latter steps were undertaken independently by two reviewers. Discrepancies were resolved through discussion or, if necessary, by consulting a third reviewer. Thirty-two quantitative papers were deemed eligible for analysis and have been reported [20]. Seventeen qualitative papers also identified during the review process and set aside for separate analysis are the focus of this manuscript.

Synthesis and quality appraisal of qualitative papers

All team members completed a review of the 17 papers to determine appropriateness for inclusion in the final analysis. Five papers were excluded because their primary objective was not to describe financial impact but rather to describe other types of challenges and impacts related to cancer (i.e. treatment-decision-making, losing a job, undergoing fertility treatment) or the work was conducted in the USA.

The 12 remaining papers were assessed for quality independently by two team members using the Critical Appraisal Skills Programme Checklist [22] (Table 1). The CASP Checklist is a 10-item tool to appraise qualitative papers

Table 1 CASP quality appraisal programme criteria

Type	CASP quality appraisal criteria
Screening (If 'yes' to both, proceed with full review)	1. Was there a clear statement of the aims of the research? 2. Is a qualitative methodology appropriate?
Methodology	3. Was the research design appropriate to address the aims of the research? 4. Was the recruitment strategy appropriate to the aims of the research? 5. Were the data collected in a way that addressed the research issue? 6. Has the relationship between researcher and survivors been adequately considered? 7. Have ethical issues been taken into consideration? 8. Was the data analysis sufficiently rigorous? 9. Is there a clear statement of findings? 10. How valuable is the research?

http://www.casp-uk.net/wp-content/uploads/2011/11/CASP-Qualitative-Research-Checklist-31.05.13.pdf

without using a numerical score and allows application of consistent review approaches. It was designed as a reminder for researchers of the importance of every criterion in qualitative papers to reduce bias (i.e. influence of researchers and a lack of transparency in data collection and analysis).

Data extraction from the 12 papers was completed by one investigator (MF). Data regarding study characteristics (aims, methodology, sampling, data collection, analysis, and interpretation) were extracted and entered into a Microsoft Excel spreadsheet. Participant quotes, categories, and themes along with key observations from the original authors' discussions and conclusions were extracted, including all original text reported in the results sections of the abstracts and manuscripts (including tables).

The process for the narrative synthesis drew on guidance from several authors [23–27]. We did not use an existing conceptual framework or model as we wished to approach the analysis without pre-existing expectations but rather see what had been reported by investigators using qualitative approaches. We focused on the experiences concerning finances as reported by patients/family members in the original manuscript presentations and did not focus on other experiences or non-financial challenges described in the papers (e.g. experience of returning to work, decision-making about insurance).

Working with the extracted data, all concepts reported in the original studies were identified, beginning with the earliest publication and building the list of concepts as the other publications were read. Once the list was complete, similarities and differences in what concepts had been explored throughout the original studies were identified. The concepts were then grouped into clusters reflecting broader constructs and collated into summary tables. The collated information was reviewed by all team members, and overarching themes were agreed upon to describe the experience of financial toxicity from the perspective of patients and informal carers. Taken together, the themes reflect a storyline or narrative of living with financial hardship associated with cancer. A conceptual map was developed to illustrate the experience and impact of the financial toxicity and summary tables were designed to provide details about the underlining concepts and themes.

Results

Summary

The 12 papers reported on 10 different studies (see Table 2). All studies were cross-sectional and utilized a qualitative descriptive design with two naming a specific type: phenomenology [36] and grounded theory [2]. Studies were conducted in four countries: two in Canada [9, 36, 37], two in Australia [28, 29], three in England [2, 30, 31], and three in Ireland [32–34] with the earliest published in 2005 [9, 28] and the most recent in 2018 [35, 37].

The labels used for the phenomenon under investigation varied. They included financial burden [2, 9, 28, 33, 34, 37], situation [2, 9], impact [30, 31], problems and issues [35], concerns [36], struggles [33], strain [28], difficulties [2], distress [33], and stress [30]; economic well-being [30], hardship [2, 32], burden [32, 34], security [2], and worries [2]; and costs [29], expense [29], expense difficulties [29], and service affordability [29].

Seven studies recruited specific populations: Aboriginal Australians [29], patients travelling for radiation treatment [28], socio-economically disadvantaged women with breast cancer [35], individuals with cancer who had accessed social welfare support [30, 31], cancer patients who had reported financial hardship [33, 34, 36, 37], breast cancer patients [9, 35], and colorectal cancer patients [32]. The remaining studies reported they sought participants purposively to maximize variation in age, gender, SES, and work histories.

Across the ten studies, 243 unique cancer patients and 56 family carers were interviewed (see Table 2). Gender, age, residence, SES level, and work history were not reported consistently. Seven studies include people with a mix of cancers (with the breakdown of cancer sites generally not specified),

Table 2 Summary of study papers ($N = 12$)

Country	Author, date, [ref]	Purpose	Method	Sample	Key findings
Australia	Hegney et al., 2005, [28]	To understand the experiences of rural-dwelling cancer patients travelling for radiation treatment	Descriptive qualitative design Individual interviews	17 rural-dwelling patients travelling for radiation treatment; Mixed cancers; 36–80 years; 14 women/3 men; Varied SES; and work histories	-Financial burden one of five themes reported -9 patients clearly identified finances were a burden -Lack of awareness about what services were available to assist with travel expenses (Note: 3 recruited to study and subsequently declined radiotherapy because of travel costs)
Australia	Shahid et al., 2011, [29]	To explore experiences and barriers of Aboriginal people in accessing cancer care services and treatment in Western Australia (explore differences based on urban versus rural-dwelling peoples)	Descriptive qualitative Individual interviews followed by member checking	30 Aboriginal cancer patients; Mix of urban ($n = 11$); and rural/remote dwelling (age, gender, employment, cancer type not reported)	- Aboriginal Australians experience profound logistical and practical barriers to accessing treatment for cancer - One of the themes regarding barriers is financial (travel, accommodation, medications, and treatment) - Need for travel and accommodation assistance was identified
England	Moffatt et al., 2010, [30]	To explore the experiences of accessing a welfare benefit service by recipients who received benefits for which they were entitled	Descriptive qualitative design Individual interviews	22 cancer patients; Mixed cancers; 35–83 years; 11 men and 11 women; 2 working; All living in high SES deprivation areas	-Financial stress compounded stress of illness -Resulting problems and distress linked to personal circumstances; can be significant and long-lasting -Managing financial burden was challenging for some patients - Accessing welfare benefits service is beneficial to cancer patients -Lack of knowledge about available welfare benefits and access to service was a significant barrier to patients
England	Amir et al., 2012, [2]	To explore the perspectives of patients and carers regarding the impact that cancer-related financial hardship/worries can have on family life	Exploratory qualitative (grounded theory)	40 cancer patient and 17 family carers; Mixed cancers; 19 patients and 7 male; Range of ages, SES levels, time since diagnosis; 16 retired/7 sick leave; 14 working FT/PT	Financial worries and difficulties can have a major impact on family lifestyle, activities, roles, and relationship Themes included: -Awareness that cancer can affect finances -The impact of cancer on family finances -The impact of finances on family lifestyle and activities -The impact of finances on family roles and responsibilities
England	Moffat et al., 2012, [31]	To evaluate a welfare benefit service from a qualitative perspective: what impact does a welfare benefit service have on the quality of life and well-being of cancer patients	Descriptive qualitative Individual interviews	35 pts (30–80) and 9 family carers Mixed cancers 19 males; 5 working FT-PT/9 retired	-Receiving benefits had an immediate and direct impact on lessening the financial burden, improving well-being, and facilitating engaging in usual activities -Assisting patients in applying for welfare benefits was effective in overcoming barriers and welcomed by recipients (Note: conceptual framework designed for illustrating impact of welfare service)
Ireland	O'Ceilleachair et al., 2012, [32]	To investigate the broad spectrum of economic and emotional consequences and their interrelationships for colorectal cancer patients and their families	Phenomenology Individual interviews	22 cancer patients and 6 family carers; 44–82 years; 17 diagnosed with colon and 5 with rectal; 14 women and 8 men; 13 lived in urban areas; 8 working	-The economic impact of colorectal cancer for patients and their families is complex and suggest that there is a strong inter-relationship between the economic and emotional impact of the disease. -Support can interrupt the negative cycle. Themes included: -Out-of-pocket costs, making ends meet, the role of family and friends, services and entitlements, and emotional costs

Table 2 (continued)

Country	Author, date, [ref]	Purpose	Method	Sample	Key findings
					(Note: model devised for the relationship between economic and emotional consequences and mediating factors)
Ireland	Timmons et al. (2 papers), 2013, [33, 34]	Paper 1: To explore the financial impact (financial adjustments and financial distress) of a cancer diagnosis and implications for financial protection in health Paper 2: To explore the financial impact of cancer in Ireland	Descriptive qualitative Individual interviews	20 patients; mixed cancers; 14 female; < 50 = 7, 50–59 = 8, 60 + = 5; 14 working at time of dx, 1 working after diagnosis 50% urban dwelling 21 social workers with oncology focus*	Paper 1: Financial distress during cancer is common and healthcare system may not provide sufficient protection Themes included: -Financial adjustments needed -Emotional consequences of financial distress -Risk factors for financial difficulties and financial distress (Note: detailed tables, reading adjustments, emotional impact, risk factors) Paper 2: The impact of the financial and economic burden of cancer on individual patients and their families is complex, multidimensional, and variable. Changes in income post-cancer exacerbate the effects of cancer-related out-of-pocket expenses. Themes included: -The wide range of cancer-related out-of-pocket costs which may be incurred -The impact on employment and income for patients and families (Note: detailed tables of costs and impacts)
Ireland	Byrne et al., 2018, [35]	To obtain information from disadvantaged Irish women concerning possible barriers to care for breast cancer above and beyond risk factors collected by cancer registries	Descriptive quantitative Chart abstraction and individual interviews	17 women with breast cancer; Average age 45 years (8 = over 50) at dx	The study isolated barriers to cancer care from the perspective of disadvantaged women. -An overarching theme was stress originating from disadvantage. -One barrier was financial. -During treatment themes included stress related to financial difficulties and caring for relatives, use of alternative therapies, difficulties accessing ancillary services
Canada	Lauzier et al., 2005, [9]	To understand breast cancer patients' and caregivers' conceptualization and sources of costs	Descriptive qualitative design Individual interview followed by focus group	26 breast cancer patients (28–78 years; 4–16 months post-diagnosis) and 24 caregivers (29–75 years); 7 in each group not working at diagnosis	Overarching themes: - Types of costs and sources of costs - Breast cancer patients and families can incur significant costs during illness experience - Resulting distress linked to individual circumstances
Canada	Longo et al., 2016, [36]; Fitch et al., 2018, [37]	Paper 1: To understand the full breadth of cancer-related out-of-pocket patient costs and lost income, private insurance behaviours, and management of finances during cancer treatment in Ontario Paper 2: To explore cancer patients' perspective about the impact of financial burden on quality of life	Descriptive qualitative Individuals interviews	14 cancer patients; Mixed cancers Age 37–78 years (average 51.4) None worked during treatment Mixed SES	Paper 1: - There are personal financial costs associated with having cancer and challenges in managing those -Many patients indicated that supplemental health insurance would have made their cancer journey less stressful. Subthemes: financial challenges and financial management, expenditures by type, residence changes, insurance value and accessibility, diet and exercise, clothing needs. Paper 2:

Table 2 (continued)

Country	Author, date, [ref]	Purpose	Method	Sample	Key findings
					-All participants experienced reduced income and increased out-of-pocket costs following their diagnosis. -Financial challenges had profound impacts on the social, psychological, and spiritual domains of quality of life from the financial challenges. -Participants worried about having sufficient funds to support themselves in the long term and felt this added to the burden they were carrying. -Most found the financial challenges surprising and unexpected

Data were extracted for patients and family carers where samples were mixed and not professional participants (i.e. social workers)

two included only breast cancer, and one colorectal cancer. For nine studies reporting gender, 146 of the 213 participants were female with 86 known to have breast cancer. Ages ranged from 30 to 89 years, some papers reporting means and other reporting categorical results. Five studies reported a mix of urban (total of 129) and rural (total of 47) dwelling patients [29–33], and one focused only on 30 rural-dwelling individuals [28].

Samples in three studies included a mix of SES levels [2, 28, 36, 37] and three focused on low-SES level patients [30, 31, 35]. Nine papers reported work status specifically [2, 9, 30–34, 36, 37]. Time since diagnosis was only reported in three papers [2, 9, 28] and stage/disease progression reported in one [35].

All studies conducted individual interviews with patients. Four included family caregivers [2, 9, 31, 32], and one included oncology-specific social workers [33, 34]. Three included an additional data collection approach: chart abstraction [35], focus groups [9], and member checking [29]. All studies worked with verbatim interview transcripts and performed content or thematic analyses. Two papers offered conceptual maps illustrating their results [31, 32].

Quality appraisal

None of the 12 papers met all CASP quality criteria (see Table 3). Five papers [32–34, 36, 37] were missing only one criterion (i.e. relationship between the researcher and participants). This was not reported in any of the other papers either. The remaining seven papers had between two and six criteria unmet. The criteria missing most frequently (three times each) were the research design appropriate for the research aims, the data collection procedures appropriate for the research aims, the data analysis sufficiently rigorous, and the implications and transferability clearly articulated. In most cases, details were missing meaning that a full assessment of the quality could not be made. The strength across the papers was the clear and detailed presentation of findings.

Narrative synthesis

The twelve papers illustrate the complexity and multifaceted nature of experiencing financial hardship following a cancer diagnosis. Each contributes to the whole picture, providing different viewpoints regarding various and diverse forms of financial hardship, the process of confronting financial challenges, working to overcome difficulties, and coping with resulting impacts. All begin with acknowledgement that facing cancer is a life-altering event and undergoing treatment is

Table 3 Results of quality assessment using Critical Appraisal Skills Programme (CASP)'s quality appraisal for selected papers

Author of paper [ref]	CASP criteria from Table 1									
	1	2	3	4	5	6	7	8	9	10
Hegney et al. 2005, [28]	√	X	X	?	?	X	√	√	√	?
Shahid et al. 2011, [29]	X	√	√	?	√	?	√	?	√	√
Moffat et al. 2010, [30]	√	√	√	√	√	?	?	?	√	√
Amir et al. 2012, [2]	√	√	√	√	√	?	√	√	?	X
Moffat et al. 2012, [31]	√	√	?	√	√	X	√	√	√	√
O'Ceilleachair et al. 2012, [32]	√	√	√	√	√	?	√	√	√	√
Timmons et al. 2013, [33]	√	√	√	√	√	?	√	√	√	√
Timmons et al. 2013a, [34]	√	√	√	√	√	?	√	√	√	√
Byrne et al. 2018, [35]	√	√	√	√	?	X	X	?	√	?
Laizier et al. 2005 [9]	?	?	?	√	?	X	√	√	√	√
Longo et al. 2016, [36]	√	√	√	√	√	X	√	√	√	√
Fitch et al. 2018, [37]	√	√	√	√	√	X	√	√	√	√

Legend: √ = yes, ? = uncertain/cannot tell/not evident, X = no

stressful. They all based their work on the assumption that facing the financial reality would likely add to the cancer burden. However, each paper focuses on slightly different aspects of the reality, documenting different viewpoints about financial hardship. All are needed to garner a holistic perspective of the lived experience.

Figure 1 shows our conceptual depiction—based on synthesis of the 12 papers—of the process of facing financial changes after a cancer diagnosis and coping with the resulting impact. Although the description below is presented in a linear fashion, the reality of living through the experience of financial toxicity is not; rather it is more circular and complex. Individuals' pre-existing circumstances, including available support from family and friends, and their responses at various points throughout the cancer experience influence the perception of financial hardship, the capacity to cope, and the subsequent impact on quality of life. Clearly, this can continue to change over time.

To understand this unfolding and changing experience, synthesis of the studies suggested five themes: household and medical costs are increased, financial resources are reduced, financial change and financial hardship vary, financial hardship has many consequences, various mitigation strategies are used.

Theme 1: Household and medical costs are increased

Cancer patients and their family members were generally surprised by the realization that there were costs for them to pay following their cancer diagnosis [2, 32, 36, 37]. Given the healthcare system where they lived, they expected all cancer care expenses would be covered either by governmental or private health plans and were unaware there could be costs they would have to cover themselves. Patients always linked the costs to the cancer diagnosis and treatment, perceiving these as ones that would not have happened for them if cancer had not occurred. Additionally, they had not anticipated expenses would continue to occur over an extended period of time, having seen cancer as acute rather than a chronic event [2]

Without exception, all 12 papers reported increased out-of-pocket costs for patients. Costs were reported in various levels of depth and with few instances of actual currency amounts. A wide range of items for which costs were incurred was

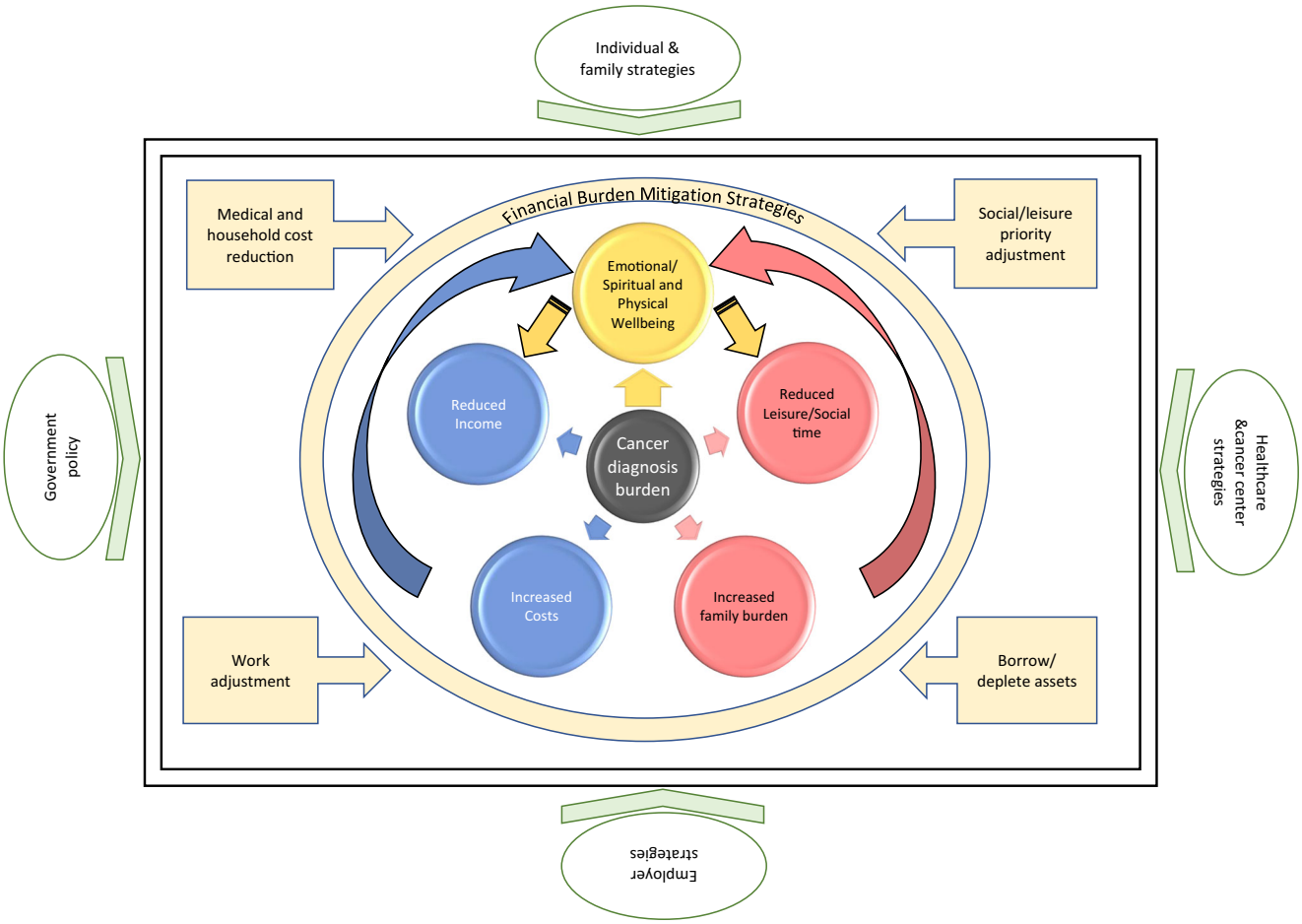

Fig. 1 Conceptual model

documented across the 12 papers (see Table 4); the specifics of which varied by individual circumstances.

Costs for travel/parking, accommodation, and food associated with hospital/clinic visits for tests, medical procedures, and treatment were cited in all 12 papers. Costs for medical supplies and medications were cited in eight and costs of tests and consultations with medical professionals mentioned in seven. Expenses not directly related to cancer treatment, but other things which changed because of the cancer situation, were reported in eight papers and included family support at home (e.g. childcare, dependent family members), household utility bills (especially heating and telephone), mortgage arrangements, and personal care. Supportive care service (e.g. counselling) costs were cited in four papers.

Theme 2: Financial resources are reduced

All 12 papers referenced that patients experienced varied levels of reduction in their income following their cancer diagnosis. Eight specifically described reduced salary income due to not working or having the capacity to only work part-time hours [2, 9, 28, 30, 31, 33, 34, 37]. Two mentioned that patients had issues with insufficient vacation coverage available [28, 32], thus having to take leave without pay. Four mentioned the amounts provided by benefit plans were insufficient to cover expenses [9, 28, 30, 31], a particularly troublesome situation for those on fixed incomes (e.g. retired). Three papers discussed reimbursements that were slow or late in coming to the patient making it difficult to meet financial commitments in the interim [30, 33, 35].

Table 4 Types/sources of costs associated with cancer (as mentioned in papers)

Category	Specific costs	Paper cited
Indirect diagnosis and treatment costs	Travel for tests/treatment (train, car, bus, plane)	2, 28–030, 31, 32, 34–36
	Travel for family member to accompany or visit patients	2, 9, 31, 32, 36
	Accommodation during testing or treatment (including relocation for duration); maintaining 2 residences	9, 31, 32, 36
	Fuel/gas for car	31, 33, 36, 37
	'Wear and tear' on car	31
	Parking	2, 9, 29, 31–34, 36
	Food—at hospital, at home to be healthy, special diet	2, 9, 29, 31–34, 36, 37
Direct diagnosis and treatment costs	Hospital bills	35, 36
	Pay outpatient/GP visits	28, 35, 36
	Pay for tests/treatments; medical consults	2, 9, 34,–36
	Pay private treatment bills	31
Medical costs—medications, devices, supplies	Paying for medications (other drugstore supplies like mouthwash, creams)	2, 9, 28, 29, 32, 35, 36,
	Homecare (professional healthcare)	9, 29, 30, 36
	Supplies–incontinence pads, dressing/bandages, special bed	2, 9, 29, 36
Supportive care expenses	Counselling (psychology) and other appointments (physiotherapist, social worker, massage therapist, etc.)	9, 35, 36
	Complementary/alternative therapies	2, 9, 36
Family care	Care of children; dependent relatives	2, 9, 30, 36, 37,
	Extra help at home (personal care, housework, yardwork, etc.)	2, 9, 29, 30, 34, 35,
Expenses related to house	Household bills (heat, telephone, etc.)	2, 9, 28, 33–37
	Mortgage	2, 28, 34, 37
	Insurance premiums	2, 29
	Adaptations/alterations to house	2, 28, 29, 34, 36
Personal expenses	Wigs/hats/scarves/make-up	2, 32, 36
	Clothing/special bras	2, 9,28, 29, 33, 34, 36
	Exercise programmes	29
	Supplements	2, 29

Theme 3: Financial changes and financial hardship vary

All papers cited financial changes had occurred for patients and families because of increased costs and reduction in available resources. However, not all individuals experienced the change as challenging or resulting in undue distress/strain/stress/hardship; individual or personal situations, context, perception, and capacity to cope influenced how much emotional distress the person felt because of the change [2, 9, 29–37].

Variation in individual circumstances [9, 33, 34, 36, 37] and variation in actual costs [32] meant some patients and families did not experience the financial changes as a burden. Some individuals were able to depend on their savings to cover the additional out-of-pocket costs and reduced financial resources [2, 9, 31–33, 37]. Others reported their health plan benefits and/or subsidies were sufficient to cover their financial expense during treatment [28, 29, 31] or they were able to depend on their partner's income [30]. One paper discussed patients' understanding about financial management and capacity to manage their financial affairs as an influencing factor in whether they experienced financial hardship [36].

However, the majority of participants in these studies experienced hardship due to financial change. Various factors were cited as placing individuals at risk (see Table 5). Having a cancer diagnosis when money is already tight is clearly a challenge [9, 30, 33–36]. Additionally, the majority of these patients indicated it was difficult to work during treatment and for various periods of time following the end of treatment, because of side effects, and they experienced reduced income or cash flow [2, 9, 28, 30, 31, 33, 34, 36, 37].

Theme 4: Financial hardship has many consequences

The impact of financial hardship was described as complex and multifaceted [34]. Clearly, it added to—and in some cases exacerbated—the emotional distress of having cancer and undergoing treatment [29–31, 33, 35, 37]. Being diagnosed with cancer was seen as a vulnerable time for individuals when the focus ought to be on dealing with the cancer and not dealing with an added burden from financial worry.

The impact of financial hardship was experienced in a number of ways [9, 34] by patients and family members [9, 34] with economic and emotional aspects closely intertwined [32–34, 37] (see Table 6). Emotional distress resulting from financial hardship was experienced to greater or lesser degrees by both patients and family members [33, 34]. Poignant descriptions of worries, anger, regret, and fears for the future were embedded in patient quotes across the papers. Some patients described feelings of remorse and concern for their family and guilt for being the cause of the financial hardship [2, 33, 34]. Some felt pressure to return to work sooner than they were ready to do so or experienced distress because family members had to go to work due to the need for income [2, 36]. For some family, relationships became troubled, and roles and responsibilities changed out of necessity [2, 37]. Fear for their future financial reality led some to grapple with existential concerns [37].

It was apparent that financial impact was not always evident in the early period of the cancer experience and only emerged later [2, 9, 30]. For some, the impact became apparent immediately (e.g. difficulty paying for transport, accommodation, food, utility bills), whereas for others, it occurred at a later point in time (e.g. paying for children's education); some costs were experienced as shorter term (e.g. transport and parking at the hospital during treatment) while others had more long-lasting implications (e.g. debt, bankruptcy, house repossession).

Lack of money was cited as a barrier to a number of cancer-related events. Individuals reported being unable to afford travel for tests to be diagnosed [35], access for medical consultation, or for treatment itself [28, 29, 33]. Lack of money meant some individuals were not able to manage their side effects well [2, 33] or were not able to obtain necessary help at home [2]. Patients perceived the lack of money was a barrier to their recovery [33] and compounded their distress [30, 37].

A few individuals reported positive impacts emerged from the financial hardship [31, 37]. Families discovered they spent more time together and engaged in less expensive, even healthier, activities [37] (e.g. walks in parks). Some reported they realigned their spending priorities and spent less on unnecessary items [31].

Theme 5: Various mitigating strategies are used

Ten papers reported strategies used to mitigate the impact of financial hardship (see Table 7). Some strategies were utilized by individuals (e.g. reduction in household spending) while others were relevant for use by cancer programmes (e.g. provide staff member to help patients with application for benefits) and policy (e.g. decrease time interval for reimbursements). Strategies were implemented at different times as the cancer experience unfolded, and awareness of financial concerns arose. Some were effective for individuals and reduced the sense of hardship (e.g. shopping at discount stores) while others were not entirely effective (e.g. going back to work before an individual was ready) and the emotional toll was heightened. The availability of strategies was a function of individual circumstances. Some strategies could be used by anyone (e.g. household budgeting) while others were not an option (e.g. using insurance, benefit plans from employers, savings accrued). Knowledge about budgeting and financial planning and skill or capacity to perform personal financial management were important attributes [36].

The negative impact of financial hardship was reduced when patients could access health plans or sick benefits or received welfare benefits as it enabled them to focus on

Table 5 Described *specifically* in papers as *risk factors* for financial hardship and/or distress associated with cancer or factors increasing financial vulnerability

Financial hardship/burden depends on personal circumstances/inter-play of factors (each case is different)		2, 9, 28–30, 32-37
Domains	Factors	
Work status	Working at time of diagnosis	2, 33, 35
	Patient is main breadwinner	33
	Self-employed	9, 28. 33, 35
	Retired	2
Location of residence	Having to travel a long distance	9
	Living rurally and no transport	33, 35
Financial status	No income, low income at diagnosis, poor, unemployed	9, 28, 33, 35–37
	Insufficient cash flow	29
	Have a mortgage	2, 33, 35
	No health insurance	9, 29, 32, 33,
	No medical card*	28, 33, 35
	Little or no savings	28, 33, 35
	Ineligible for benefits	33, 35
Personal circumstances	Having children	2. 33–35
	Living alone	33, 35
	Being a single parent	33, 35
	Being younger	35
	Being older	35
	Having several other medical conditions	9
Family support	Lack of/little family/friend support	28, 33-35
	Primary carer/partner becomes ill	28
Financial literacy	Lack of financial planning/knowledge	29
	Inability to manage money	29

*Provides access free at the point of delivery to public health system services in Ireland (i.e. no co-payments)

dealing with the disease [30, 31]. Practical or tangible aid had a direct and immediate benefit in helping individuals pay household bills, eat more healthily, and afford treatment costs, thus reducing worry and stress [30, 31]. Financial aid provided a safety net [30] and preserved savings [32]. Some participates reported an easing of worry by just receiving financial advice without tangible aid being provided [30].

Accessing benefits was reported as a challenge for many who considered this strategy. Patients were not always aware of benefit plans or coverage available to them [28–32, 34] or reported they were not told about them by healthcare professionals [30, 31]. Upon investigating, some found out they were ineligible for particular plans/subsidies/benefits [30–34, 36] or the premiums are too costly [36]. Others reported it was hard to pursue the application on their own [28, 30, 31] or when they were feeling so unwell (fatigue, nauseas, etc.) [30, 31]. Some participants found it hard to talk with healthcare providers about finances or found healthcare providers insensitive [30, 32]. Many reported difficulties in actually asking for financial assistance because they felt overwhelmed, intimidated, embarrassed, and proud or they held negative attitudes about seeking benefits/claims for illness purposes [2, 29–32, 35]. Many also found the paperwork to apply for assistance complicated [29, 31, 34]. Individuals who had had help completing application forms reported this was invaluable [30, 31].

Recommendations from papers

Several papers made recommendations aimed toward offsetting financial hardship for cancer patients and their families. Recommendations for patients included encouragement to learn about the potential for financial consequences and review one's personal financial status early during the cancer experience [2]. Recommendations for employers emphasized being aware of their employees' situation when cancer strikes and facilitating time off work, flexible working hours and arrangements, and access to sick leave benefits [32–34, 36]. Specific suggestions included not forcing employees to take unpaid leaves or use vacation time [34] and facilitating 'early' retirement without penalty if the employee wishes to retire [33].

Recommendations for healthcare professionals included being aware of the potential for patients to suffer financial

Table 6 Impact on individual/family of financial hardship/burden associated with cancer as described by participants in papers (items directly impacted or at risk for being impacted)

Domain	Specific impact of financial hardship described	Paper
Family finances	Depletion of savings	30, 35
	Loss economic self-reliance	29
	Unable to care for dependent relatives	2
	Not pay for children's education/university	35
	Not have money for retirement	35
	Not be able to pay back money borrowed	35
	Not pay mortgage	34, 35
	Not pay household bills (heat, telephone)	34, 35
	Reduced eligibility for insurance	35
	Unanticipated one-time costs/purchases—e.g. bed, home alterations	2, 28, 29, 34, 36
	May bring about debt, bankruptcy, house repossession	2, 9,
Lifestyle: social/leisure activities	Decreased in holidays/going away	2, 30
	Decreased visiting with friends/family	2, 30, 34
	Decreased eating out	2, 34,
	Decreased cable TV/network	2
	Decreased theatre, movies, concerts, sports attendance	2, 30
	Decreased 'treats' for children; gifts	2
Psychological well-being	Reduced sense of financial security	29, 30, 37,
	Range of emotions to various degrees for both patient and family members: worry, anxiety, fear for future, remorse, guilt, regret, anger, disappointments	2, 9, 28–30, 33–36
	Reduced sense of self-esteem, self-worth	30
Health	Less healthy eating	2, 30
	Forego treatments	31, 32
	Forego medications	32, 36
Spiritual well-being	Distress/uncertainty regarding future-life priorities	30
Family dynamics	Change in roles and responsibilities—relationships	2, 36
	Decreased social and leisure activities together, decreased holiday/vacation time	30, 34, 37
	Patient may move in to live with children	30
	Family member may have to work, add hours	30, 33
	Family members may have to stop working, or reduce hours, take unpaid leave, earn less income	2, 30, 33

hardship [2, 29] and acting proactively to screen and assess for financial challenges early in the course of the cancer experience [2, 30, 31]. Clinicians ought to have knowledge about resources available for financial assistance and refer patients appropriately [2, 30, 31, 34].

At cancer programme/centre levels, recommendations were made about organizing multiple avenues for patients and families to obtain advice about financial assistance [29, 30] and be able to easily self-refer for help [30]. Specific strategies included implementing a navigator role for financial assistance [29, 35], providing practical help with benefits applications [30, 31], integrating a benefits welfare organization into the cancer programme [30], and providing free transportation and accommodation programmes [29, 32, 34]. Finally, recommendations for benefits programmes included reviewing eligibility criteria [33] and simplifying application process/forms [29].

Discussion

This systematic narrative review of qualitative studies regarding financial toxicity provides, for the first time, a holistic picture about the experiences of confronting financial toxicity following a cancer diagnosis. This work offers a set of ideas about important issues/factors which influence financial hardship for cancer patients and their families and strategies to mitigate some of the factors. Across the four countries represented—all of which had publicly funded healthcare systems where financial protections in health might be expected

Table 7 Mentioned as *intentional/purposely selected* strategies by individuals use to cope with/manage financial difficulties associated with cancer

Domain	Strategy	Paper
Manage healthcare	Ration medications	35
	Forego treatment	35
	Adjust treatment schedule	28
Manage household spending	Decrease spending budget	9, 33, 35
	Economize on household bills/spending	33, 35
	Scrimp/save and juggle spending; 'rob Peter to pay Paul'	35
	Turn down heat	35
	Decrease food shopping; stretch food;	35
	Buy secondhand clothes/discount stores/cheaper brands	30, 32, 33, 35
Adjust social or leisure activities	Decrease social and leisure activities (not eat out; not visit friends/family; no concerts, theatre, sports events, etc.)	30, 33, 35
	Find inexpensive things to do	30
	Decrease hairdresser appointments	35
	Cut back or do not take holidays/vacations	30, 35
	Cut back on children's activities (sports, clubs, etc.)	30, 35
	Work hard/try to continue with children 'as normal'; scrimp for self so children can have music, sports, camps, school, university, etc.	2, 33, 35
Borrow	Borrow from banks/credit union	35
	Borrow from employer	31,35
	Access charitable/welfare organization for support	32-34, 37
	Borrow from family/depend on family for transport, groceries	9, 28, 30 31, 32, 35, 36
Adjust work schedule	Work extra shifts (family member)	2, 33
	Family member takes time off work (to care for patient)	2, 33
	Go back to work as soon as possible/early	2, 35
	Pay using credit card	32
Deplete resources	Access savings	28, 30, 35, 36
	Have mortgage protection plan	33
	Negotiate with bank (more favourable rates); refinance home	35,37
	Have partial coverage from benefits package/insurance	9, 32
	Have insurance coverage	29, 33
	Dependence on high income partner	33
Adjust priorities	Change priorities/trade-offs/redefine priorities	9, 30
	Change retirement plans	30
	Abandon planned projects	9
	Sell possessions	2
	Seek out benefits package/insurance	29, 33–35

to be stronger—there were consistent issues regarding travel, food, accommodation, and family support that pinpoint gaps which institutions and employers need to consider. This synthesis suggests where action could be taken to start closing these gaps.

Clearly, financial change can become a reality for many cancer patients following their diagnosis and can have a broad spectrum of impacts not only for the individual but also for the family in terms of hardship and emotional and psychological well-being which are over and above the direct impacts on health and well-being from the disease itself. There is a dynamic nature to the situation as financial change becomes real and individuals begin to cope with the impacts and initiate mitigating strategies. The magnitude of the hardship varies and not all patients find it highly distressing and hard to cope financially. Pre-diagnostic financial circumstances (income, savings, assets) are clearly important here as well as the supports that are available within the system. As has been reported elsewhere, the extent of cancer-related OOPC incurred tells us nothing about an individual's ability to pay [38]; indeed,

there is an association between OOPC and ability to pay—those with greater ability to meet costs incur more costs [39]. Results here advance understanding by demonstrating that the financial burden of cancer is complex and multidimensional. We need to do more than measure OOPC if we want to understand financial burden of cancer—just focusing on OOPC may be misleading and fail to identify those in greatest need.

Additionally, it is important to note the financial consequences ripple out from the person with cancer through the family. Family members also feel practical and emotional impacts and are likely to be highly involved with implementing mitigating strategies. Further investigation is needed to identify the true indirect burden on family members and friends; there has been relatively little focus on this in the literature to date. These observations also have implications for assessment, intervention, and support of the family regarding financial concerns, likely delivered in tandem to exchanges with the patient.

It is also of interest to note from our work that the burden of mitigation strategies seems to fall rather heavily upon the individual and the household (e.g. reduced work time, saving depletion, increased expenditure out of income) and how little of the burden seems to be alleviated by external bodies (e.g. government welfare, and insurers). This raises questions about who is responsible for acting to offset the issues and the interaction between the health and social welfare safety nets within such settings.

The observations have implications for risk assessment at the onset of treatment, early during the person's experience, and sharing of information about the potential for financial toxicity and available resources to help offset the issue. They also imply a need to monitor the emotional distress and financial toxicity on an ongoing basis. Embedding financial information in survivorship care plans has been suggested [40]. Future work could focus on designing a clinically relevant, robust, and brief risk assessment tool for healthcare professionals to apply at point of care in busy clinical environments [41].

There is also a need for developing and evaluating relevant interventions to ameliorate financial burden, and to maintain financial and emotional well-being and QoL. QoL is increasingly recognized as an independent prognostic indicator in cancer [42]; therefore, attention should be paid to issues which adversely impact QoL. Some strategies could appropriately be directed to the individual at point of care (e.g. providing advice on managing finances, building financial literacy) while others would be more appropriately implemented at cancer programme/clinic levels (e.g. support to navigate social welfare system/make applications, drive to care schemes). At the very least, cancer programmes could make it a priority to have information about financial resources for cancer patients readily available.

We set out to study experiences of cancer patients in settings with publicly funded healthcare systems. It might be expected that such systems would afford citizens some degree of financial protection in health. It is therefore interesting and important to note that financial toxicity also occurs in highly developed countries with publicly funded healthcare systems. It would be interesting to compare the observations from these studies with qualitative results from other settings where the healthcare system is predominately private, and in countries that are not categorized as having a very high human development index. This literature is beginning to emerge [43–45]. The few US-based qualitative studies report similar patient concerns but emphasize different priorities regarding mitigating strategies [46, 47]. Patients in other countries may experience different types or magnitudes of financial changes and hardships or the perceived nature of the hardship could vary.

Clearly, additional research on this topic using qualitative approaches would be valuable. A notable proportion of participants in the eligible studies were women with breast cancer who are likely to have higher education/economic status and very likely different work patterns than men. Focus is needed on other cancer populations, particularly those cancers where incidence is highest in people of lower SES (e.g. lung, head, and neck) who might be expected to have fewer 'resources' at the time of diagnosis. Additionally, research to date has included primarily Caucasian populations. Research with other ethnic and cultural groups is important as these groups may experience even more significant (or different) impacts. It would also be useful if common language and definitions were devised for use in studying the phenomenon of financial burden and distress following a cancer diagnosis.

Additionally, much of the research has been cross-sectional. Longitudinal qualitative studies would be valuable to understand how experience evolves over time and better understand how people cope, adjust, and manage. Studies of experiences of patients receiving newer targeted/biological therapies, which are particularly costly and have different side effect profiles in contrast to traditional therapies, would also be important to complete.

Limitations

Although we searched multiple databases, we did not search grey literature; therefore, this possibly cannot be excluded that we missed relevant data. We also restricted consideration to papers published in English. In terms of limitations of the evidence-based, it was striking that the 12 papers came from only four counties, all of which are English-speaking.

Conclusions

Consistent gaps in support across four countries highlight that strategies to mitigate financial effects related to travel, accommodation, medications, family support, and income

replacement are still needed for a significant portion of the populations in these countries. Cancer patients and their families can experience a broad range of impacts when they are facing and coping with financial toxicity. Identification of those at risk for financial toxicity early during the cancer experience is important. Mitigation strategies need to be designed for both point of care and system application, and our investigation seems to suggest that these strategies are needed in all four countries observed.

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Article review for quality: MF, LS, PH
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Declarations

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